

Arnaldo Silva

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EDUCATION

CLARKSON UNIVERSITY

Master of Science

Major in Electrical Engineering

Cumulative GPA: 3.93/4.0

Relevant Coursework: Strategic Project Management, Power System Analysis I/II

Potsdam, NY

Dec 2020 – Jun 2022

SUFFOLK UNIVERSITY

Bachelor of Science in Engineering

Major in Electrical and Computer Engineering

Relevant Coursework: CS, Digital Electronics, Circuit Theory I/II, Linear Systems, Solid State Devices, Digital Signal Processing, Linear Systems, Control Systems, Embedded Systems, Data and Computer Communications

Boston, MA

Aug 2015 – May 2019

WORK EXPERIENCE

NATIONAL GRID

Reliability Analytics Engineer

- Led data-driven initiatives reducing SAIFI by 35% and SAIDI by 10%, preventing penalties.

- Automated workflows via Python, SQL, PowerBI, cutting report times by 90% and doubling team capacity and visualizing KPIs.

- Enhanced grid performance by identifying outage patterns, reducing problem circuits by 40%.

Associate Reliability Analytics Engineer

- Completed Graduate Development Program (GDP) and led capstone reducing outage durations down to minimum of 5%.

- Managed EV Central, boosting charger availability to 90% and proving a 40% emission reduction.

- Handled regulatory filings for reliability metrics (SAIDI, SAIFI, CAIDI).

Intern

- Optimized solar panel efficiency (+15%) through tilt/azimuth testing.
- Spearheaded customer billing insight project, enhancing user experience.
- Researched and advocated for advanced metering infrastructure.

APPLE

Technical Expert / Technical Specialist

- Diagnosed and repaired Apple devices, maintaining a 90%+ NPS score.
- Trained team members and escalated engineering issues.

Previous Roles: Apple Master at Sprint and Best Buy (2013 – 2017)

Northborough, MA

Sep 2020 – Present

Jun 2019 – Sep 2020

Jun 2019 – Sep 2020

Chestnut Hill, MA

Nov 2017 – Apr 2022

UNIVERSITY PROJECTS

SMART SHOES

2019

- Designed wearable health-tracking prototype integrating sensors and microcontrollers, communicating data via Bluetooth coded using C++.

PEER-TO-PEER IP COMMUNICATION

2018

- Built encrypted messaging/file transfer system in C++.

MORSE DECODER

2017

- Developed Raspberry Pi-based Morse code interpreter

ADDITIONAL SKILLS

Hardware Engineering: PCB Layout (KiCAD), Microcontroller Programming, Circuit Design, Verilog/VHDL

Software & Tools: MATLAB/Simulink, Multisim, Microsoft Office Suite, Lab Equipment (Oscilloscope/Multimeter)

Programming: C++, Python, Java, Swift, SQL, Assembly

Data & Automation: SQL, Python, PowerBI, Snowflake, Excel Macros

Languages: English, Spanish

Certifications & Training: Fundamentals of Engineering (Electrical & Computer Engineer-In-Training), Advanced Certificate in Power Systems Engineering

Awards: Nominated for “Big Work” at National Grid